INF2006 Team Project
Occuscope
Group: G007

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1. Problem, Users & Success Criteria

1.1 The Problem

Students at SIT Punggol still walk through the campus to judge how busy a floor is, whether a discussion room is free, and what bookings are made that day. Occuscope is a campus map (campus → building → floor → location) that shows Quiet / Moderate / Crowded, today's events, and discussion-room Booked / Available in Singapore time. The interface is a crowd heatmap plus a 3D campus view; the browser reads occupancy from the API.

Users

Primary: SIT students choosing where to study and, once signed in, booking discussion rooms.

Secondary: visitors and other campus users who only need the public heatmap and today's events.

1.2 The Scope

The map covers 64 Punggol spaces (teaching blocks, library, Foodgle Hub, Wholesome) for AY2026/27 Trimester 1 teaching hours (08:00–20:00). Occupancy is generated from identified NUS ROBOD occupant counts and transferred onto SIT rooms by type, hour, weekday, the published academic calendar, and SIT capacity. Crowd bands are computed on the server: Quiet ≤ 30%, Moderate ≤ 70%, Crowded otherwise. NUS Wi-Fi floor counts were inspected and left out of v0. Login and discussion-room booking are product features on that map. The 3D view follows SIT's Campus Wayfinder so buildings are recognisable.

1.3 Success Criteria

Product

• A student can browse building → floor → room and read Quiet / Moderate / Crowded for a chosen Singapore hour in AY2026/27 Trimester 1 (08:00–20:00).
• Discussion rooms show Booked or Available (08:00–20:00 SGT) together with occupancy; study spaces and food courts show occupancy only.
• The UI labels occupancy as estimated, while the README and report explain how the values are generated.
• Signed-in students can book discussion rooms in 30-minute blocks under a 240 minutes/week cap, with cancellation.
• The service is deployed serverless on AWS Academy, with dated evidence for function, security, data/AI, and scale.

Data / AI

• NUS hold-out MAE uses a leakage-safe date split.
• v0 is the lowest-MAE model on that split, including when the simple baseline wins.
• Each map hour is stored with source = generated.
• The heatmap uses occupancy_ratio / crowd_level at a chosen instant (?at=). The last seed timestamp is trimester break and would look empty.
• Each location has a next-two-hour prediction from the generated series (model_version =

v0).
• Dated data/AI evidence: evidence/test-data-ai.md (objective, setup, commands, actual, artefacts).
• Training files stay in the analytics pipeline; they are outside git and outside occuscope.db.

2. Solution Overview & Architecture

Occuscope is a same-origin web app: the campus map and booking UI talk only to a HTTP API. Occupancy is trained offline on NUS ROBOD, with an optional Wi-Fi ablation, then seeded into the application database.

Two Stores:
1. Training: data/raw/ ROBOD CSVs + raw_data.xlsx (gitignored). Analytics only.
2. Application: building / location / occupancy / event / occupancy_prediction, plus demo students and room_booking. Locally, SQLite (data/occuscope.db); on AWS, the same file is a private S3 object. Seeded by python src/db/init_app_db.py from data/sample/.

Deployed Stack:
Public cloud, us-east-1, AWS Academy Learner Lab. API Gateway HTTP API crowdmap-http fronts three Python 3.11 Lambdas: crowdmap-web (frontend from a private S3 web bucket), crowdmap-api (map, forecasts, events, chat), crowdmap-bookings (login and bookings; reserved concurrency 1). CloudWatch (7 alarms + dashboard crowdmap) and SNS crowdmap-alerts provide observability; AWS Budget crowdmap-monthly sends monthly cost alerts. Groq hosts the Octopus chatbot; occupancy stays precomputed in SQLite.

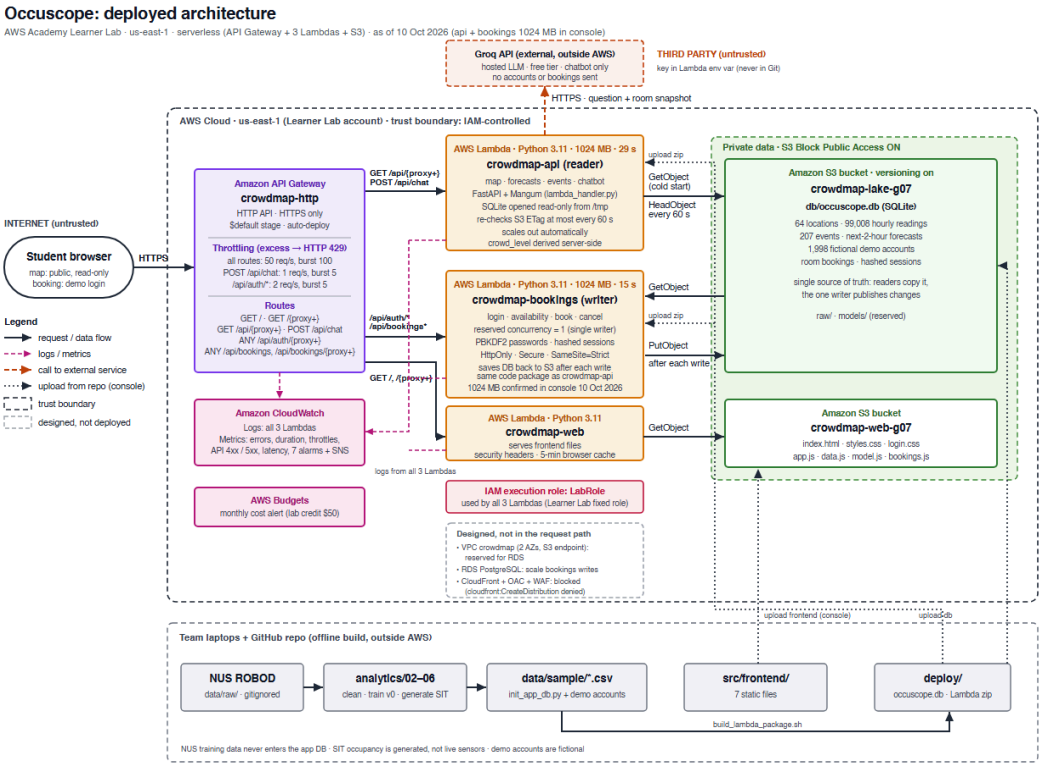


Figure 1. Occuscope on AWS Academy Learner Lab (us-east-1).

Labels: ***crowdmap-http*** *(API Gateway),* ***crowdmap-web****,* ***crowdmap-api****,* ***crowdmap-bookings*** *(Lambda), private buckets* ***crowdmap-web-g07*** *(site files) and* ***crowdmap-lake-g07*** *(SQLite). Groq sits outside AWS. RDS is shown dashed as a production option.*

Trust Boundaries and Data Flows:
1. Browser → HTTPS API Gateway. Occupancy GET is public campus information. **Login** needs the custom header **X-Occuscope-Request: 1**. **Bookings and cancellations** need that header **plus** a session cookie. **Chat** is **POST /api/chat** on **crowdmap-api** (no session guard; read-only advice).
2. The frontend consumes **map_x**, **map_y**, **occupancy_ratio** and **crowd_level** from **GET /occupancy/current?at=**.
3. **crowdmap-api** reads SQLite copied from private S3 (ETag refresh about 60 s). Forecasts are precomputed rows.
4. **crowdmap-bookings** is the only writer for **login/bookings**; it uploads the database after a successful write. Chat POST stays on **crowdmap-api**.
5. Chat → Groq with an allowlisted room snapshot; each call is request-scoped.
6. Analytics run offline; generated CSVs are seeded into SQLite.

Component Responsibilities:
- Occupancy generation and metrics → Lideon
- REST API, schema, auth/booking/chat APIs → Zul
- Heatmap + 3D wayfinder UI, login/booking screens, Octopus UI → Zi Qian
- AWS deploy / scale → Kristen
- Security tests and evidence pack → Ryan

3. Cloud Service, Deployment Choices & Trade-Offs

Deployment model:
Public cloud, single region (us-east-1), AWS Academy Learner Lab. No on-premises part. Users reach the service on the public internet.

Service model:
Serverless FaaS with managed PaaS around it. The IaaS/PaaS boundary sits below our code. AWS manages TLS, servers, patching and scale-out; the team owns code, configuration, data and access.

Layer	**Service**	**AWS**	**Team**
Entry	API Gateway crowdmap-http	TLS, availability, throttle engine	Routes, throttle limits
Compute	Lambda crowdmap-web, crowdmap-api, crowdmap-bookings (Python 3.11)	OS, runtime, scale-out	Code, memory, timeout, concurrency, env
Storage	Private S3 web + lake	Durability, encryption at rest	Block Public Access, versioning, layout

Layer	Service	AWS	Team
Operations	CloudWatch, SNS, Budgets	Collection and delivery	Alarm thresholds, who is notified
Chat	Groq (external)	Groq hosts and operates the model service	API key, request handling and POST /api/chat throttling

Alternatives considered
EC2 + ALB + Auto Scaling (IaaS) — about USD 30/month idle before traffic; the team would own OS patching. Too much of a USD 50 lab credit.
ECS on Fargate + ALB — still billed per hour; extra image/registry steps unused for a small Python API.
Amazon RDS — the production path for concurrent writes (about USD 12–15/month db.t3.micro, Lambda in VPC). The running demo uses the single-writer Lambda. A VPC (2 AZs, private subnets, S3 gateway, no NAT) is already provisioned for RDS.
CloudFront + private S3 — preferred (CDN, WAF). Learner Lab denied CloudFront, so crowdmap-web Lambda serves the private bucket over HTTPS.
S3 static website hosting — would make the web bucket public. Rejected.

Assumptions and constraints:
Learner Lab: shared LabRole, CloudFront blocked, sessions expire (voc-cancel-cred), USD 50 credit (evidence/deploy-log.md). Occupancy is generated; 1,998 demo accounts. Expected workload: about 2,000 daily users × about 40 API calls ≈ 80,000 requests/day, short peaks between classes. Cost controls are in section 7. Lambda runs Python 3.11; occupancy is precomputed, so inference stays off the request path.

4. Implementation, Data Design & Security Controls

4.1 Frontend

Dependency-free HTML/CSS/JavaScript (src/frontend/). The Node dev server proxies /api and serves an allowlisted static set. Students search and filter rooms, use an original 3D campus model aligned to the SIT wayfinder, and see Booked / Available separately from occupancy. Foodgle Hub (E4 L1) and Wholesome (W3 L2) are occupancy-only. Login, 30-minute booking blocks, weekly allowance, grouped cancellation, and the Octopus chat control are in the UI.

4.2 Backend

FastAPI implements src/api-contract.md against src/db/schema.sql. Timestamps are ISO-8601 with +08:00. Crowd bands are derived on the server from occupancy_count / capacity (Quiet ≤ 0.30, Moderate ≤ 0.70, else Crowded).

Persistent data: buildings, locations, occupancy (source dummy / generated / model), events, demo app_user / credentials / sessions, and room_booking. Locally

data/occuscope.db; on AWS the same file is the S3 lake object. Seed from data/sample/ (DATA_DICTIONARY.md).

Map GET is public. **GET /occupancy/current** without **at** returns 422. Timelines max seven days. **GET /occupancy/{id}/prediction?at=** returns the next two available occupancy entries after the selected instant from the precomputed series. The frontend displays only entries within the following two hours.

Mutating routes: email/password login (PBKDF2; **X-Occuscope-Request: 1**) and discussion-room bookings (240 min/week, 30-minute slots 08:00–20:00 SGT, atomic create and grouped cancel; header plus session cookie). Booked / Available comes from the booking table. 1,998 demo accounts with a student-email format check. Chat is POST /api/chat on crowdmap-api.

On AWS, map/forecast/event **reads** and **chat POST** go to **crowdmap-api**. **Login and bookings** go to the single-writer **crowdmap-bookings**.

4.3 Cloud

See section 3 for service choice. Implementation: src/backend/lambda_handler.py (reader Lambdas + single-writer S3 sync), src/infra/build_lambda_package.sh, API Gateway throttling, seven CloudWatch alarms, tests/load_test.py. Deployment record: evidence/deploy-log.md.

4.4 Data & Machine Learning Pipeline

Scripts (repo root, Python 3.11.9 for v0):

• analytics/02_clean_robod.py → data/processed/robod_clean.csv (52,128 rows; HVAC dropped)
• analytics/03_eda_robod.py → analytics/figures/
• analytics/04_train.py → metrics_holdout.csv, occupancy_v0.joblib (local, gitignored)
• analytics/05_generate_sit.py → occupancy_generated.csv, occupancy_prediction.csv
• analytics/06_holdout_diagnostics.py → MAE-by-type/hour, 7 vs 14-day window, persist vs v0, NUS crowd-band counts
• src/db/init_app_db.py rebuilds the app DB from sample CSVs, demo students, and discussion-room demo bookings (240 min/week, no same-student overlap)

Pins: analytics/requirements.txt (pandas 3.0.6, scikit-learn 1.9.1, joblib 1.6.0, …). random_state=42.

SIT rooms: compiled from RBS catalogue screenshots. E2, E6, W3 DR02–DR17, W5 DR18–DR20 and DR23–DR26 (the catalogue has no DR21/DR22). Discussion-room capacity from furniture (about 8). W1 library open study and meeting rooms; E2/E6 wayfinder study spaces; Foodgle Hub E4 L1 and Wholesome W3 L2. 64 locations. ACE seminar rooms stay on the taught-room side of the catalogue.

4.5 Security Controls

Named threat: forged Quiet / Moderate / Crowded. The API derives **crowd_level** from people ÷ capacity (quiet ≤ 30%, moderate ≤ 70%, else crowded). Extra query parameters such as **crowd_level=crowded** leave the JSON unchanged (**tests/test_api_security.py**, **14/14 tests**, 10 October 2026, **evidence/security-api-local.txt**). Injection-style ids return 404. Occupancy GET opens SQLite read-only.

Campus occupancy GET is public. **Login** needs **X-Occuscope-Request: 1**. **Bookings and cancel** need that header plus a session cookie. **Chat** is **POST /api/chat** on **crowdmap-api** without that session guard. Unsigned login/booking POSTs returned **403** on the 10 Oct lab smokes. Auth throttle 2 req/s.

Secrets: .env ignored; **.env.example** placeholders; hygiene and secret-scan tests. Raw NUS files stay out of git and out of **occuscope.db**. Map: **evidence/threat-control-map.md**.

Learner Lab uses shared LabRole (no custom IAM JSON export). Buckets: Block Public Access. Static review: 8 open findings (7 Low, 1 Medium); SR-6 closed. Dev-server test 6/6. Live URLs and account IDs stay out of the repository.

5. Analytics & AI/ML Feature: Data, Method, Evaluation & Limitation

The assessed data/AI feature is the occupancy series that drives Quiet / Moderate / Crowded on the map and the two-hour lookahead. Octopus is a separate Groq call. v0 is trained locally; hours are precomputed into the database.

5.1 Question

Which time and room-type model achieves the lowest MAE on held-out NUS ROBOD dates, and how can its output be adapted into a calendar-aware occupancy series for the SIT map?

5.2 Data Used

ROBOD combined_Room1–5.csv: 7 September 2021 – 23 December 2021, 5-minute ticks, +08:00. R1–R2 lecture, R3–R4 office, R5 library. Weekday extract. Target: occupant_count. After clean: 52,128 rows (Tekler et al., Building Simulation 2022; Figshare 10.6084/m9.figshare.19234530).

ROBOD's room-level Wi-Fi device counts have a correlation of approximately 0.69 with its occupant counts. A separate NUS floor Wi-Fi dataset was examined in an ablation using hour and weekday profiles (Zenodo DOI: 10.5281/zenodo.17578240; CC BY 4.0; sheet 5min, levels L2–L6). Neither Wi-Fi input is used by the selected v0 model because SIT has no matching feed.

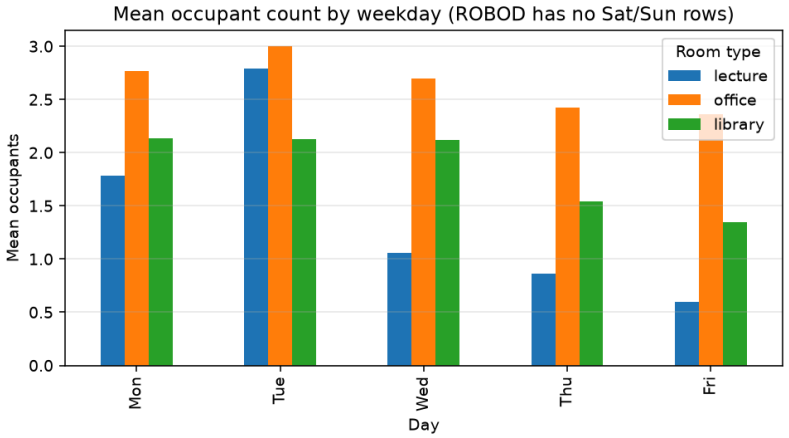


Figure 2. ROBOD by weekday. Monday to Friday extract.

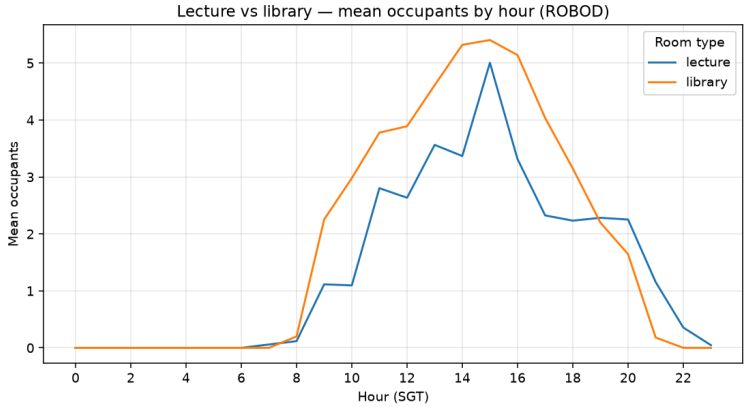


Figure 3. Lecture rooms vs library. Room type is a feature because the two behave differently.

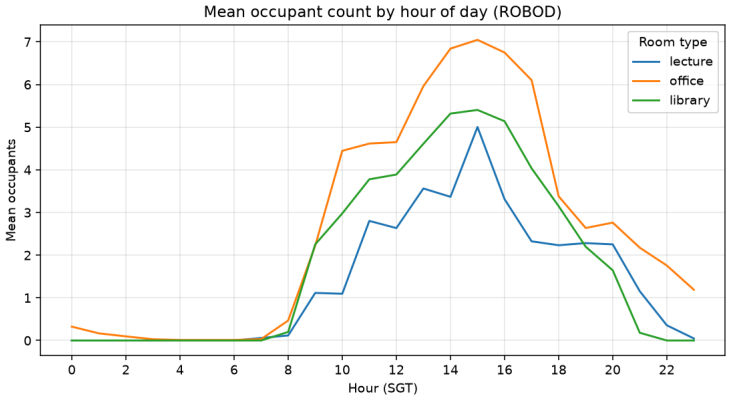


Figure 4. ROBOD occupancy by hour of day (24 September 2026 EDA). Peak about 15:00 SGT; overnight empty so overall means look small.

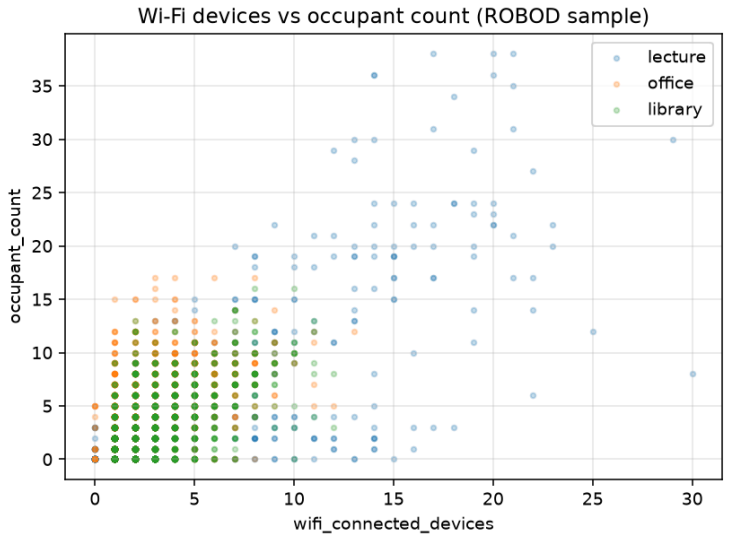


Figure 5. ROBOD room-level Wi-Fi device counts versus occupant counts (correlation approximately 0.69). Wi-Fi was evaluated in an ablation but is excluded from v0.

5.3 Method

Features transferable to SIT: hour, day_of_week, room_type. v0 omits Wi-Fi and CO2.
Split: last 14 distinct dates held out (15 November 2021 – 23 December 2021). Train 34,560 rows; test 17,568.
Models compared: hour × room-type mean (baseline), Ridge, Random Forest (random_state=42).

SIT generate (unlabelled transfer):
• Type map: discussion_room → office; library / study → library (occupancy_model.py).
• Utilisation = v0 count / ROBOD type 95th percentile, clip, × SIT capacity × calendar × mixed hashed event turnout × per-room mix.
• Overlay sit_calendar.json: AY2026/27 Tri 1 (31 August – 27 December 2026), MOM 2026 holidays, E = IT IWSP mix, W3/W5 = other courses, W1 library shared. Event titles are independent of turnout. OIP is off (2026 dates unpublished). Weeks 12–14 raise discussion-room booking chance and study occupancy.
• Timestamps ISO-8601 +08:00. Hourly 08:00–20:00. 64 locations; 99,008 generated rows.
• Predictions: next two generated hours after the selected instant (GET /occupancy/{id}/prediction?at=).

5.4 Results (NUS Hold-Out)

Model	MAE	RMSE	R^2
Hour x Room-Type Mean (Selected)	1.74	2.92	-0.04
Ridge	1.83	2.86	0.01
Random Forest	1.84	3.06	-0.14

The hour × room-type mean was selected because it achieved the lowest MAE. Its negative R^2 means it performed worse under squared error than predicting the held-out target mean. Differences between the training and late-term hold-out periods may contribute to this result.

Lecture MAE 2.04 vs library 1.60 / office 1.56; 15:00 MAE 4.93. A 7-day window MAE is 1.73. Persist (copy last hour) beats v0 on NUS 1–2 h, but it needs a last reading, which SIT rooms do not have; the map therefore uses v0. Room Wi-Fi lowers NUS MAE to about 1.31 (Ridge); it is an ablation only and is absent from the app DB.

SIT example (Wednesday 30 September 2026, 15:00 SGT), generated occupancy:

Location	**Count**	**Cap**	**Ratio**	**Band**
E2-04-20-DR223	4	8	0.50	moderate
W3-03-07-DR02	4	8	0.50	moderate
E2-03-07-DR209	0	8	0	quiet
W3-03-10-DR03	6	8	0.75	crowded
W1-04-OPEN (library open study)	9	80	0.1125	quiet

The teaching-hour demo uses **?at=2026-09-30T15:00:00+08:00**. The last seed row is trimester break (empty). Overlay check: Deepavali-in-lieu (9 November) lowers counts versus the next teaching Monday. Full dated record: **evidence/test-data-ai.md**. DR223 is the functional-test room (4/8, booked that hour, source=generated).

5.5 Limitations (data / AI)

1. Model performance is evaluated on held-out NUS ROBOD observations. The generated SIT values are checked against capacities, calendar rules and application behaviour. Accuracy against real SIT occupancy was estimated.
2. This ROBOD extract is weekdays only; weekend map hours use an explicit calendar factor.
3. Wi-Fi counts associated devices. They were an ablation only.
4. Calendar and IWSP mix use published dates plus a programme assumption.
5. Capacities come from furniture photos; the catalogue from screenshots.
6. The map displays generated occupancy estimates for wayfinding. The “Estimated occupancy” label does not imply access to live SIT sensors.

6. Testing, Scalability & Monitoring Results

Four tests. Objective, setup, command, expected/actual, date and artefact path live in the evidence files.

#	Test	Evidence	Date	Result
1	Functional workflow	evidence/test-functional.md	10 Oct 2026, Ryan; ZQ + Zul chat sign-off	Browser 10-step pass. DR223 at 15:00 SGT 30 Sep: **4/8 moderate, generated, booked**. 64 locations.
2	Security (named threat)	evidence/test-security.md	10 Oct 2026, Ryan	API **14/14 tests**. Extra **crowd_level** leaves JSON unchanged. Login needs header; bookings need session + header.
3	Data / AI	evidence/test-data-ai.md	24 Sep–10 Oct 2026, Lideon	v0 MAE **1.74**; SIT occupancy generated.
4	Scale / resilience	evidence/test-resilience.md	8 Oct 2026, Kristen	CloudShell load test. Reads scale; writer serialises; spike returns 429, 0 × 5xx.

Implementation and measurements. Lambdas scale out per request. API Gateway sheds excess (50 req/s stage; chat 1; auth 2). Bookings go through one writer (reserved concurrency 1) that uploads SQLite to S3; readers refresh on ETag (~60 s). /api/health reports role=reader on the map Lambda. S3 versioning is the recovery path.

Phase	What it showed
A steady (10 clients, 60 s)	All 200. At 128 MB, p50 **1,582 ms** (CPU-bound). After raising memory, p50 **206 ms**, ~5.9 → **48.7 req/s**.
B spike (60 clients)	1,122 × 200, **3,150 × 429**, **0 × 5xx**. Throttling kept the map up.
C auth burst	401 / 429 as designed; **7 × 503** when writes overlapped the single writer.
D booking after cold start	After retries: **PASS** in 3.9 s (persistence-run.txt).

Reads scale with 0 × 5xx on the map spike. The writer keeps bookings; overlapping writes retry. The production write path is RDS; the demo uses the single writer. Ryan's 10 Oct console check: crowdmap-api and crowdmap-bookings both 1024 MB / 29 s and 15 s. Phase D was left at the earlier memory. crowdmap-web remained 128 MB.

Monitoring (evidence/monitoring.md, written 10 Oct from 8 Oct artefacts; CloudWatch was left at those exports). Seven alarms email via SNS crowdmap-alerts: Lambda Errors ≥ 1 on each of three functions; bookings Throttles ≥ 1; API 5xx ≥ 5, 4xx ≥ 50, p95 ≥ 3 s (5 min); dashboard crowdmap.

During the spike, the **4xx** alarm was mostly HTTP 429 (2,962 4xx in five minutes). **5xx** and **bookings-throttled** both showed **7**, matching phase C's seven 503s. Email in about five minutes; 4xx and 5xx returned to OK without a manual reset.

7. Cost, Sustainability, & Operational Considerations

Cost controls. Pay-per-request only (deployment check: 0 EC2 / RDS / ALB / NAT). S3 gateway endpoint instead of NAT (~USD 32/month). Throttling caps abuse. Budget crowdmap-monthly. The load-test cost excludes chat.

Estimated monthly cost at ~2.4 million requests (before free tier): API Gateway ~2.40, Lambda requests ~0.48, Lambda compute ~8.40, S3 < 0.50, 7 alarms 0.70 → **about USD 12.50**. EC2 + ALB would be ~USD 30/month idle.

Cost risk: each write uploads ~23 MB; versions kept → ~11 GB/day at 500 bookings/day (~USD 8/month growing). Recommended mitigation: lifecycle delete noncurrent versions after 1 day. Status: recommended, still unconfirmed on the lab account. RDS would remove this path in production.

Sustainability. Scale to zero at night and holidays. 1024 MB made map requests about 8× faster at similar GB-seconds. Next efficiency step: cache /occupancy/current per chosen hour.

Operations. src/infra/build_lambda_package.sh; verify_deployment.sh; check_versions.sh (**14/14 MATCH** on 8 October). Memory settings were lost on a re-upload until the check; infrastructure-as-code is the main operational improvement. Groq key is a Lambda env var on **crowdmap-api** (CLI **--query** so values stay off the screen).

8. Team Contributions, Ethical Considerations & Reflection

Ethics / responsible AI. The interface labels occupancy as estimated. The README and this report disclose that the values are generated from NUS-derived patterns and calendar assumptions, rather than live SIT sensor measurements. v0 uses public NUS ROBOD; Wi-Fi was an ablation. Login uses 1,998 fictional student accounts. The room list comes from public / RBS screenshots; demo event titles are original. Chat sends an allowlisted snapshot to Groq; transcripts are request-scoped; the UI asks users to omit personal details. The repository holds no production credentials.

Name	Role	Artefacts	Reflection
Zi Qian	Frontend / Map	Dashboard, 3D campus model, occupancy/booking UI, Login/account UI, Octopus UI	Integrated occupancy, catalogue and booking-rule updates; the model is illustrative, not surveyed.

Name	Role	Artefacts	Reflection
Zul	Backend / Database	Occupancy/events API, student auth, booking APIs (240 min/week, grouped cancel), chatbot route	HTTP contract plus write path for demo login/bookings; chat cannot book.
Lideon	Data / ML	ROBOD train/generate, calendar overlay, 64-location seed, demo booking quota, evidence/test-data-ai.md	Leakage-safe date hold-out; shipped the hour×type mean when it beat Ridge/RF. SIT values are generated and labelled as such.
Kristen	Cloud / Scalability	Serverless deploy, single-writer, throttling, alarms, load test, architecture diagram	CloudFront blocked → private bucket via Lambda. Honest single-writer limit. Memory check cut p50 from 1.6 s to 206 ms. Next time: IaC.
Ryan	Security / Testing / Monitoring	Threat–control map, named-threat tests, secret/preflight scans, static review, functional pack, monitoring write-up	I treated occupancy as public campus information and bookings as the thing that actually needs a login. The named threat did not change when POST routes landed: the browser still cannot pick Quiet/Moderate/Crowded; the API derives crowd_level from count and capacity. I would not ship my 30 September “GET-only” draft. Monitoring uses Kristen’s 8 October alarms: the 4xx spike was throttling, the seven 5xx were the single writer, and both returned to OK on their own.

References

Tekler, Z. D., et al. (2022). ROBOD. Building Simulation.
https://doi.org/10.1007/s12273-022-0925-9

Figshare 10.6084/m9.figshare.19234530 ; https://github.com/ideas-lab-nus/robod
Wang / DeST Lab. NUS Wi-Fi. Zenodo 10.5281/zenodo.17578240 (CC BY 4.0)

SIT AY2026/27 academic calendar:
https://www.singaporetech.edu.sg/admissions/undergraduate/academic-calendar-sit-and-joint-programmes

MOM public holidays 2026:
https://www.mom.gov.sg/newsroom/press-releases/2025/0616-public-holidays-for-2026

SIT Punggol Campus (Foodgle / Wholesome hours):
https://www.singaporetech.edu.sg/about/punggol-campus

SIT Campus Wayfinder (visual reference only):
https://www.singaporetech.edu.sg/campus-wayfinder

Appendix A - AI Usage

Copy of **AI_USE_DECLARATION.md**.

Tool	Where used	What it produced	Team verification	Licence / source
Cursor (Grok)	Repo kickoff, 23 September 2026	Manifest skeleton, folder layout, sample SIT CSVs, SQLite schema/init, EDA stub, README / data dictionary / API contract	Lideon reviewed against the INF2006 brief and inspected ROBOD / Zenodo columns locally	N/A (AI-assisted authoring)
Cursor (Grok)	Occupancy v0 and SIT generate, 26 September 2026	Training script, hold-out metrics, generation overlay from the public SIT calendar, README / dictionary / data-AI evidence updates	Lideon compared models on NUS hold-out, checked generated ratios against the academic calendar, and retained the hour×type mean as v0	N/A (AI-assisted authoring)
Cursor (Grok)	Security evidence, 26 September 2026	**evidence/threat-control-map.md**, **evidence/test-security.md**, **evidence/monitoring.md**; gitignore unittest; network intended control	Ryan checked against the INF2006 brief (threat-map path, test headings, secrets handling, no live-sensor claim)	N/A (AI-assisted authoring)
Cursor (Grok)	Tri 1 seed + store hygiene, 29 September 2026	Full-trimester generate, MOM holidays, API **?at=** notes, **tests/test_data_store_hygiene.py**	Lideon checked timestamps (**+08:00**), generated sources in SQLite, and that NUS raw stays out of git and the app DB	N/A (AI-assisted authoring)
Cursor (Grok)	RBS rooms + Wi-Fi ablation, 30 September 2026	Catalogue locations; **metrics_wifi_ablation.csv**; crowd/seed tests; pinned **analytics/requirements.txt**	Lideon matched room ids to RBS screenshots and kept v0 without Wi-Fi after NUS MAE improved with room Wi-Fi	N/A (AI-assisted authoring)
Codex	Frontend and original model, 29–30 September 2026	**src/frontend/**: UI, data adapter, development proxy, tests and documentation	Agent ran five data/API tests and checked room/floor selection in-browser; human team review pending. Original procedural campus model; catalogue-driven floors and markers.	AI-assisted UI and original illustrative geometry; SIT wayfinder used as reference
Cursor	Security evidence, 30 September 2026	**tests/test_api_security.py**, evidence updates	Ryan ran the tests and reviewed outputs against **src/api-contract.md**	N/A
Cursor	Functional evidence, 30 September 2026	**evidence/test-functional.md**, **evidence/functional-run-local.txt**, manifest run section	Ryan ran the commands and checked outputs against	N/A

Tool	Where used	What it produced	Team verification	Licence / source
			src/api-contract.md and **src/frontend/README.md**	
Cursor	Preflight and secret scan, 30 September 2026	**tests/test_no_secrets_in_tracked_files.py**, **tests/test_submission_preflight.py**, monitoring export commands, report draft (local)	Ryan ran the tests and reviewed the outputs; report draft to be checked against evidence	N/A
Cursor	Static security review, 30 September 2026	**evidence/security-review.md**, **tests/test_frontend_server_security.py**, report-draft check	Ryan verified each finding against the cited lines and ran the test	N/A
Cursor	Local security pack, 10 October 2026	**tests/test_api_security.py** auth/booking supporting tests; **evidence/test-security.md**, **threat-control-map.md**, **security-review.md**; **.env.example** Groq placeholder; click-through checklist in **src/frontend/README.md** and **evidence/test-functional.md**	Ryan ran the 14 API security tests and the offline suites; browser checklist was executed 10 Oct	N/A
Claude (Anthropic, Cowork)	Cloud deployment, 25 Sep – 8 Oct 2026	Step-by-step AWS console/CLI guidance; drafts of **src/backend/lambda_handler.py**, **src/infra/*.sh**, **make_expected_versions.py**, **tests/load_test.py**, **evidence/architecture.svg/png**, **evidence/deploy-log.md**, **evidence/test-resilience.md**, cloud report section drafts (kept local; not in the submission ZIP)	Kristen ran every command herself in AWS CloudShell / console, checked live results (deployment check, 14/14 version match, alarm emails, load-test output), and corrected the drafts against the real outputs; failed runs are reported as they happened	N/A (AI-assisted authoring)

Datasets and baselines: ROBOD — Tekler et al., Building Simulation 2022; Figshare DOI 10.6084/m9.figshare.19234530; GitHub ideas-lab-nus/robod.

NUS Wi-Fi floor counts — Wang / DeST Lab, Zenodo DOI 10.5281/zenodo.17578240, CC BY 4.0.

SIT Campus Wayfinder used as a visual reference only; Occuscope renders its own illustrative model.

Appendix B - Contributions

Copy of **TEAM_CONTRIBUTIONS.md**.

Name, Student ID	**Role**	**Artefacts**	**Test / evidence ownership**	**Reflection**
Zi Qian, 2501958	Frontend / Map	**src/frontend/** dashboard, original 3D campus model, occupancy/history/forecast/ event views, Booked vs occupancy cards, Login/account/booking UI, Octopus chat UI; **evidence/test-frontend-map.md**	Functional / map workflow (with Ryan's **evidence/test-functional.md**)	Integrated occupancy, catalogue, and booking-rule updates from teammates into the site, including catching demo bookings that broke the 240 min/week and overlap rules. The campus model is independently authored and illustrative, not a surveyed architectural plan. Occuscope login and reservations are project demo features, not official SIT systems.
Zul, 2500993	Backend / Database	**src/backend/** occupancy/events API against **src/api-contract.md**; student auth (hashed passwords, sessions, login/logout); discussion-room booking APIs (availability, create, mine, grouped cancel, 240 min/week); Groq chatbot route; booking schema	Functional API evidence (with Ryan); **src/backend/test_bookings.py**, **test_cancel_group.py**, **test_chat.py**	Built the HTTP contract the map consumes, then the write path for fictional student login and discussion-room bookings, including weekly allowance, conflicts, and atomic grouped cancellation. Chat is a provider call with an allowlisted campus snapshot; it cannot book rooms. Production-scale concurrent writes still need a shared database (RDS), which Kristen documented on AWS.
Lideon, 2501098	Data / Machine Learning	**analytics/** (clean, EDA, train, generate, hold-out diagnostics); **data/** dictionary and SIT seeds (64 locations, Tri 1 occupancy, mixed event turnout); **src/db/schema.sql**, **init_app_db.py**, **demo_bookings.py** (240 min/week, no overlap); **evidence/test-data-ai.md**	Data / AI test (**evidence/test-data-ai.md**); **tests/test_crowd_and_seed.py**	Compared a leakage-safe data hold-out and shipped the hour×room-type mean when it beat Ridge/RF, rather than a more complex model. SIT map values are generated and labelled as such. Demo discussion-room seeds follow the same 240 min/week cap as the booking API.
Kristen, 2501481	Cloud / Scalability	AWS serverless deploy (API Gateway **crowdmap-http**, 3 Lambdas, private S3, VPC, budget) in **evidence/deploy-log.md**; **src/backend/lambda_handler.py** (reader / single-writer S3 sync); **src/infra/*.sh**;	Scale / resilience (**evidence/test-resilience.md**, **resilience-run.txt**, **persistence-run.txt**,	CloudFront was blocked in Learner Lab, so the frontend is served from a private bucket through Lambda instead of a public bucket. Bookings needed writes, which SQLite on Lambda cannot share across instances, so a single-writer

Name, Student ID	**Role**	**Artefacts**	**Test / evidence ownership**	**Reflection**
		throttling, 7 CloudWatch alarms + SNS + dashboard; **tests/load_test.py**; **evidence/architecture.png**	**steady-1024mb .txt**); **evidence/aws-deployment-check.txt**, **version-check.txt**; **evidence/alarm-notifications.txt**	Lambda was used and its limit recorded honestly. Map p50 was 1.6 s at 128 MB; raising memory cut p50 to 206 ms. Next time: infrastructure-as-code so settings cannot be lost on re-upload. Groq key rotation and IAM JSON were not re-exported.
Ryan, 2501205	Security / Monitoring / Testing	**evidence/threat-control-map.md**; named-threat **evidence/test-security.md**; **evidence/security-review.md**; **tests/test_gitignore_secrets.py**, **test_api_security.py**, **test_no_secrets_in_tracked_files.py**, **test_submission_preflight.py**, **test_frontend_server_security.py**; **evidence/test-functional.md**; **evidence/monitoring.md**; manifest **run** section; files the four-test pack	**evidence/test-security.md**; coordinates the four tests; **evidence/monitoring.md** (8 Oct alarms, interpretation 10 Oct)	I treated occupancy as public and bookings as the thing that actually needs a login. The named threat stays the same: the browser cannot pick Quiet/Moderate/Crowded; the API derives **crowd_level** from count and capacity. I would not ship the 30 September GET-only write-up as-is, because login and booking POSTs are real now, with hashed passwords, HttpOnly cookies and a custom request header. Monitoring is Kristen's 8 October alarms: the 4xx spike was throttling, the seven 5xx were the single writer, and both returned to OK on their own.